

CRIME IN INDIA

Crime in Indian Cities (2020-2024)

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Github: <https://github.com/prayag-pds/crime-india-power-bi.git>

Tools Used

Cleaning & Analysis: **Power BI**

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INTRODUCTION-

This report examines a comprehensive dataset of criminal activities across multiple Indian cities from 2020 to 2024. The data details crime types, timing, weapon use, victim profiles, and police deployment, covering offenses from identity theft to homicide. It also tracks case closure rates, offering valuable insights for understanding crime patterns and improving public safety strategies. The Project was conducted with Power BI.

DATA DESCRIPTION-

- **Source:** Indian Crimes Dataset → [Kaggle](#) → [Licence](#)
- **Time Frame:** 2020-2024
- **Size:** 40,000+
- **Variables:** Crime type, date/time, weapon used, victim demographics, police deployment, case status

DATA PREPARATION (ETL Process)-

- Extracted data from Kaggle and loaded it to Power BI.
- Initially, there was an issue with the 'Date of occurrence' column because it was in a different format. Using Power BI's Transform tab, I rectified the errors by applying the appropriate Locale to change the language/region.
- Extracted date from "Date Reported" and added "Time Reported".
- Extracted the date in "Date of Occurrence" as there was already a "Time of occurrence" column.
- In the "Victim Gender" column, convert M to Male and F to Female, and kept "X" as is.

- There were some anomalies in the crime domain, e.g. Homicide was in other crimes rather than violent crimes, so rectified those by adding a calculated column using DAX.

```
Crime Domain New =
SWITCH(
    'India Crime'[Crime Description],
    "IDENTITY THEFT", "Other Crime",
    "HOMICIDE", "Violent Crime",
    "KIDNAPPING", "Violent Crime",
    'India Crime'[Crime Domain])
```

- Extracted Months and Years from different dates.

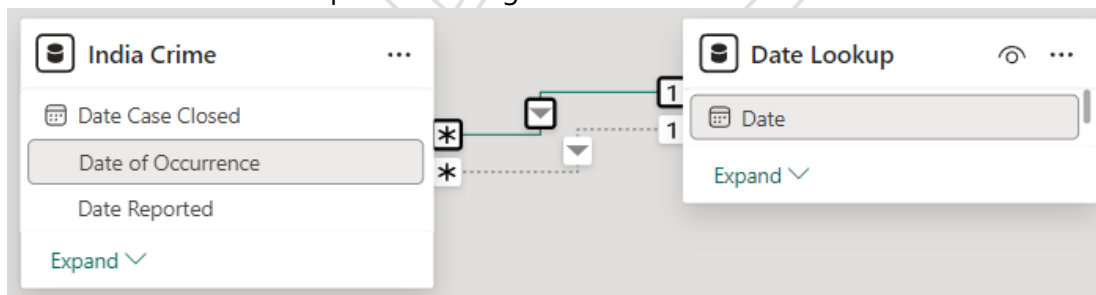
EXPLORATORY DATA ANALYSIS:

- Initially there was issue with visualization as there were gaps between dates (Many similar Dates & gaps in between dates). Using DAX added a new table – “Date Lookup”:

```
Date Lookup =
ADDCOLUMNS (
    CALENDAR (DATE(2020,1,1), DATE(2025,12,31)),
    "Year", YEAR([Date]),
    "Month Number", MONTH([Date]),
    "Month Name", FORMAT([Date], "MMMM"),
    "Year-Month", FORMAT([Date], "YYYY-MM")
)
```

Date	Year	Month Number	Month Name	Year-Month	Start of Month
01-01-2020	2020	1	January	2020-01	01-01-2020
02-01-2020	2020	1	January	2020-01	01-01-2020
03-01-2020	2020	1	January	2020-01	01-01-2020
04-01-2020	2020	1	January	2020-01	01-01-2020
05-01-2020	2020	1	January	2020-01	01-01-2020

And created a relationship with the original Crime table.



- Using DAX created Measures.
- Total Crimes committed:

```
Total Crimes =
COUNT('India Crime'[Report Number])
```

- Last Month Crimes:

```
Last Month Crimes =  
CALCULATE(  
    [Total Crimes],  
    DATEADD(  
        'Date Lookup'[Date], -1, MONTH  
    )  
)
```

- Month on Month Crime Percentage Change:

```
MoM Crime % Change =  
DIVIDE([Total Crimes] - [Last Month Crimes], [Last Month Crimes])
```

- Unsolved Crimes:

```
Unsolved Crimes =  
COUNTBLANK('India Crime'[Date Case Closed])
```

- Unsolved Crime Percentage:

```
Unsolved Crime % =  
DIVIDE([Unsolved Crimes], [Total Crimes])
```

- All Crimes:

```
All Crimes =  
CALCULATE(  
    [Total Crimes],  
    ALL('India Crime'[Report Number])  
)
```

- Case Duration in Days(Calculated Column):

```
Case Duration Days =  
DATEDIFF(  
    'India Crime'[Date Reported],  
    'India Crime'[Date Case Closed],  
    DAY  
)
```

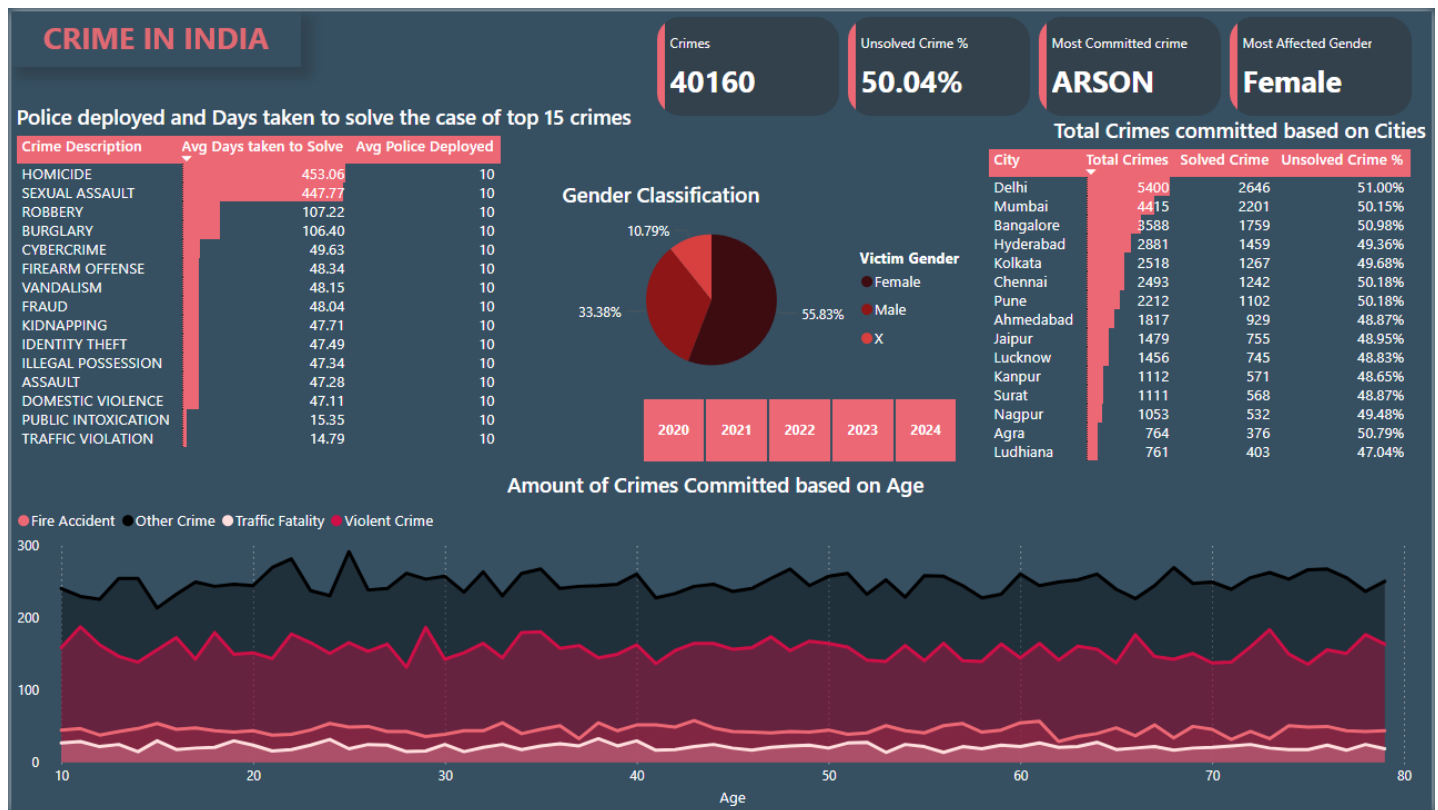
- Average Days taken to Solve:

```
Avg Days taken to Solve =  
AVERAGE (  
    'India Crime'[Case Duration Days]  
)
```

- Average Police Deployed:

```
Avg Police Deployed =  
AVERAGE (  
    'India Crime'[Police Deployed]  
)
```

VISUALIZATION-



CONCLUSION-

After doing necessary analysis and the visualization, it was found that:

- The overall **crimes committed** were **40,160** and out of these almost **50.04%** cases are **unsolved**.
- Out of these "**Arson**" was the **most committed offence**.
- Females** are the **most affected** gender, accounting for 55.83% of victims, followed by male at 33.38% and other gender identities having 10.79%.
- After other related crimes violent crimes are the most committed offence. Fire mishaps are still lower than violent crimes. After these traffic fatalities are the lowest.
- Delhi** reports the **highest crimes** contributing **5400** of total, followed by **Mumbai** with **4415**, and then **Bangalore** having **3588** of total.
- Most cities have an **unsolved crime rate** close to **50%**, with Delhi slightly higher at 51 % and other major cities like Bangalore, Chennai, and Delhi just above 50%.
- In every top city, solved crimes are fewer than half of the total, reflecting a consistent challenge in crime resolution.
- Police deployments numbers don't account to case solves**, as in the viz, through all the crimes, deployments are similar, yet there are significant differences in the duration it takes to solve. **On average**, the deployment consists of **10 police** officers, but it can take up to **450 days** on average to solve cases of '**Homicide**' and '**Sexual Assault**'. In contrast, the **same number of officers** typically solves **other** types of **crimes** within **50 days**. Therefore, the **time required for resolution really depends on the type of crime**.